

RA8P1 E-READER

Processor schematic development
Power networks are being implemented from RA8P1-specific references.
This revision is not a completed schematic or a fabrication release.

RA8P1 IO allocation



File: mcu_interfaces.kicad_sch

Global supplies: +3V3_MCU and GND.
Core regulator output MCU_VCORE stays local to the processor power sheet.

Remaining subsystem circuits follow epic #821 and issues #824-#834:
radio, memory/storage, USB/battery, system rails, display controller,
panel power, touch, warm/cool lighting, buttons and sensors.
No empty subsystem sheets are represented as completed circuits.

Footprint and PCB qualification are deferred by scope.

RA8P1 internal core regulator



File: mcu_core_power.kicad_sch

RA8P1 IO and analog supplies



File: mcu_supply_power.kicad_sch

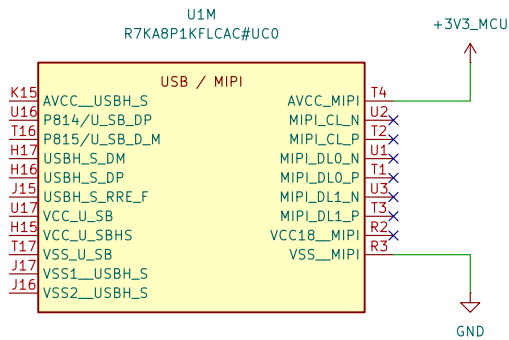
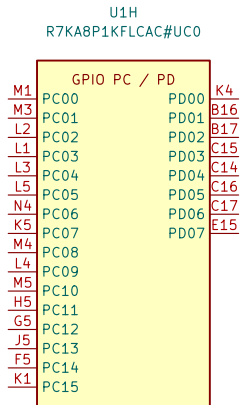
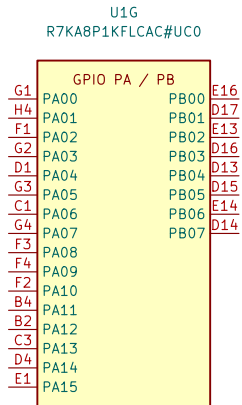
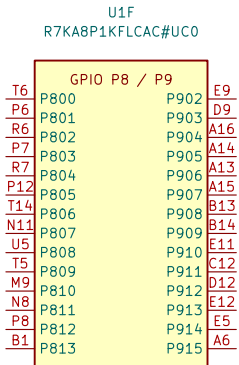
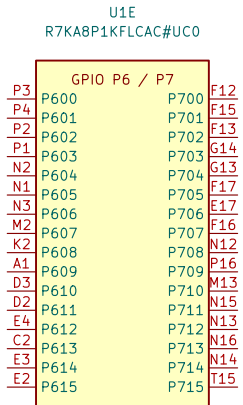
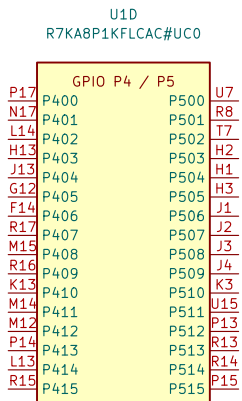
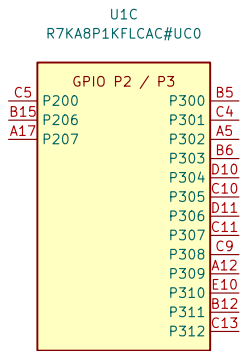
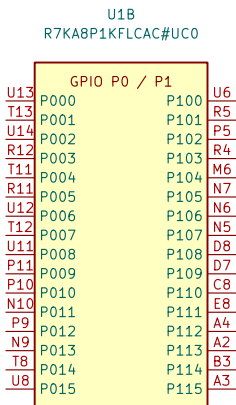
RA8P1 clocks, reset and debug



File: mcu_clocks_debug.kicad_sch

Sheet: / File: ereader_rev1.kicad_sch		
Title: RA8P1 e-reader - schematic index		
Size: A3	Date: 2026-09-05	Rev: 0.3 DRAFT
KiCad E.D.A. 10.0.5	Id: 1/5	

DRAFT: Existing processor units retained. Signal allocation and clocks/debug wiring are not yet complete.



MIPI unused: AVCC_MIPI to VCC; VSS_MIPI to GND.
Leave VCC18_MIPI and all six D-PHY lanes open.
Firmware: MSTPCRC.MSTPC10=1 and MSTPC17=1.
RA8P1 HUM Rev.1.30, section 21.4, p.862.

RA8P1 e-reader

Sheet: /RA8P1 IO allocation/
File: mcu_interfaces.kicad_sch

Title: RA8P1 IO allocation

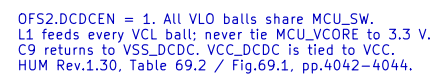
Size: A3

Date: 2026-09-05

Rev: 0.3 DRAFT

KiCad E.D.A. 10.0.5

Id: 2/5



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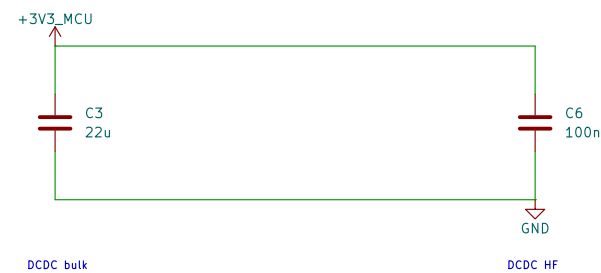
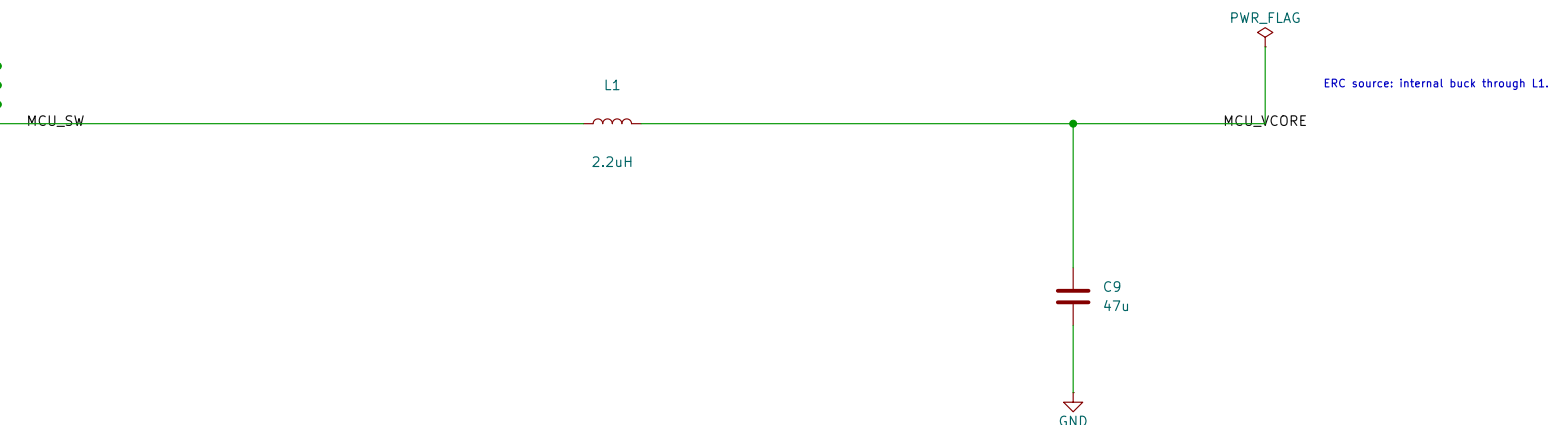
+3V3_MCU

DC/DC

A10 VCC_DCDC VLO A7
A11 VCC_DCDC VLO A8
B10 VCC_DCDC VLO B7
B11 VCC_DCDC VLO B8

A9 VSS_DCDC
B9 VSS_DCDC

GND



L1 DCR <= 100mOhm. Effective capacitance and inductor current rating require final MPN qualification. The 3.3 V source and return will be declared on the system-power sheet; no blanket ERC flags.

